PALEOMAGNETIC EVIDENCE FOR PLEISTOCENE CLOCKWISE ROTATION IN THE OISO HILLS: A POSSIBLE RECORD OF INTERACTION BETWEEN THE PHILIPPINE SEA PLATE AND NORTHEAST JAPAN

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Abstract. A paleomagnetic study was made on the Pleistocene strata in the Oiso Hills located along the boundary between the Philippine Sea plate and the northeast Japan arc. This study evaluates the sense and amount of the Quaternary deformation associated with the interaction between the plates. About 170 sedimentary rock samples ranging in age from 0.2 to 1.0 Ma were measured. We used alternating field and thermal demagnetization to examine the stability of the natural remanent magnetizations. The mean field directions from the Numashiro, the Ninomiya, and the Haneo Formations, which range in age from 0.2 to 0.7 Ma, are nearly aligned with the present axial geocentric dipole field. In contrast, the directions from the Maekawa Formation, which ranges in age from 0.8 to 1.0 Ma, show on average 50° clockwise deflections of declination. These deflections are probably caused by clockwise tectonic rotation between 0.6 and 0.9 Ma. We propose and discuss three models of the rotation mechanism, (1) local drag along a major strike-slip fault, (2) regional dextral shear along the plate boundary, and (3) oroclinal bending caused by collision of a buoyant terrane.

Introduction

It is widely accepted that the Izu block, which is located on the northern tip of the Philippine Sea (PHS) plate (Figs. 1 and 2), collided with Honshu, Japan, during the early Quaternary [e.g. Matsuda, 1978; Kitazato, 1986; Koyama, 1986a] and has been accreting into it [Tonouchi and Kobayashi, 1983; Taira et al., 1988]. Moreover, a change in relative convergence direction between the PHS plate and Honshu also occurred in the early to middle Pleistocene [Nakamura et al., 1984; Kaizuka, 1984; Seno, 1987]. If these are the case, it is expected that the areas along the boundary between the PHS plate and Honshu have been largely deformed since the early Quaternary.

Nakamura et al. [1984] proposed that convergent boundaries between plates should be divided into two categories, the material boundary and the mechanical boundary zone, when one consider in detail the tectonism on and near the surface of subduction zones. The material boundary is a

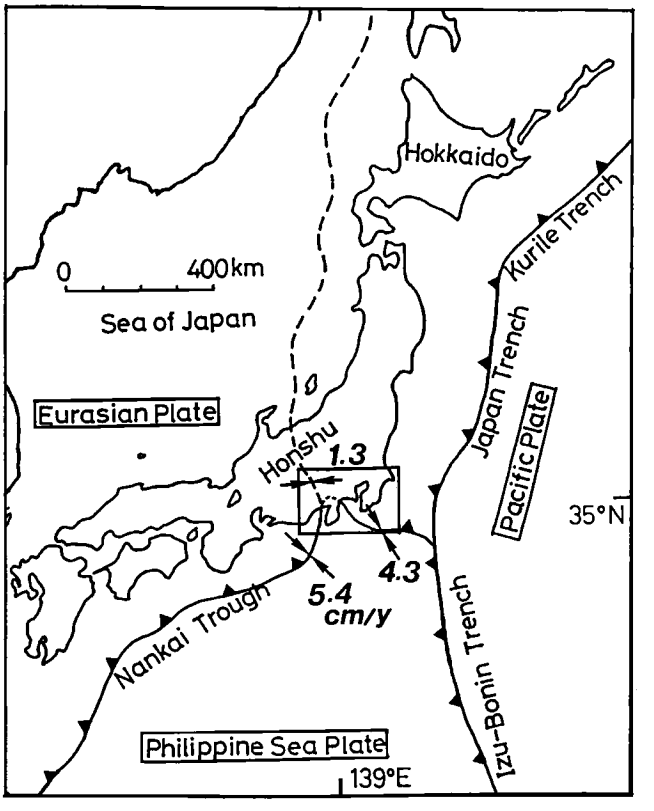


Fig. 1. Index map showing the plate arrangement around Japan. Rectangle: study region shown on Figure 2. Arrow and number: present convergence direction and rate between plates around the study region according to Minster and Jordan [1978, 1979] [Ishibashi, 1984]. Broken line: estimated convergent boundary between the Eurasian and the North American plates [Nakamura, 1983] since 0.5 Ma [Seno, 1985].

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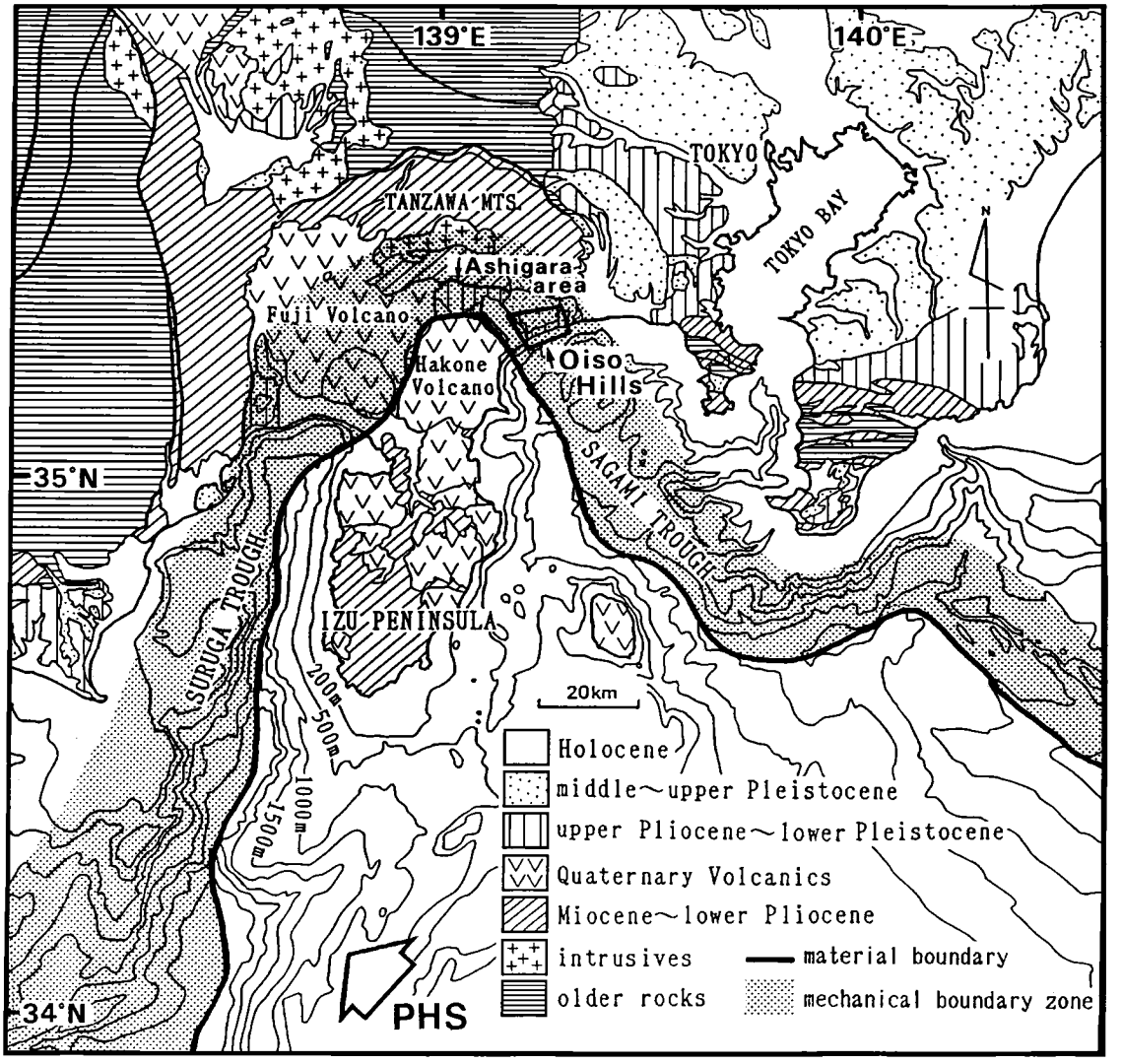


Fig. 2. Geologic sketch map of the Oiso Hills and adjacent areas modified from Geological Survey of Japan [1978] and Japan Association of Quaternary Research [1987]. Thick solid line and shaded belt: material boundary and mechanical boundary zone between the Philippine Sea plate and Honshu, respectively, after Nakamura et al. [1984]. Large arrow: estimated convergence direction between the Philippine Sea plate and northeast Japan in the early Quaternary before 0.5 Ma [Seno, 1985]. Solid frame: study area in the southern part of the Oiso Hills shown on Fig. 4.

boundary where the materials originating from two different plates are in contact with each other, and the mechanical boundary zone is a belt where the relative motion between plates is mechanically accomodated [Nakamura et al., 1984]. The locations of these two boundaries between the PHS plate and Japan in and around the study area were delineated by Nakamura et al. [1984] (Fig. 2). The Oiso Hills are located in the mechanical boundary zone and many active faults are distributed around the Hills [Kaneko, 1971; Kano et al., 1984; Ito et al., 1987]. The Kozu-Matsuda fault [Uesugi et al., 1982], which is one of the large active faults in the mechanical boundary zone [Nakamura et al., 1984], cuts the southwestern margin of the Oiso Hills.

We expect that the Oiso Hills are a suitable place for evaluating the interaction within the mechanical boundary zone between the PHS plate and northeast Japan. The present paleomagnetic study was made to detect the sense and amount of the Quaternary deformation in the Oiso Hills and to

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clarify the history of interactions between the plates.

Geology

Miocene to Holocene volcanics, volcaniclastic rocks, and marine sediments are distributed in the Oiso Hills [Association for Kanto Quaternary Research, 1987]. Early to middle Pleistocene marine sediments are mainly distributed in the southern part and are divided into the following four units: the Maekawa, the Haneo, the Ninomiya and the Numashiro Formations, in ascending order [Yano, 1986, Figs. 3 and 4]. Their ages were estimated from calcareous nannofossil biostratigraphy [Yano, 1986; Okada, 1987]. Explanations about these formations are summarized below after Yano [1986].

The Maekawa, Haneo, Ninomiya and Numashiro Formations are composed of sandy siltstones and silty sandstones intercalated with many tuff beds, and unconformably overlie the Takatoriyama Formation and the Miocene basement (Fig. 3). The Ninomiya Formation includes the Kozu Conglomerate Member and the Myoken Sandstone Member, which are a deltaic conglomerate body and a sandstone-dominated part, respectively (Fig. 3). The Numashiro Formation also includes a deltaic conglomerate part named the Fudosan Conglomerate Member (Fig. 3). The siltstones or sandstones of these formations contain many molluscan, foraminiferal, and calcareous nannoplankton fossils. The ages of the nannofossils are estimated to be 0.8-1.0 Ma for the Maekawa Formation, 0.5-0.8 Ma for the Haneo Formation, 0.3-0.5 Ma for the Ninomiya Formation, and 0.1-0.3 Ma for the Numashiro Formation, on the basis of the revised biochronology of Pleistocene calcareous nannofossils in the subtropic western Pacific [Okada, 1987].

The Ninomiya and the Numashiro Formations unconformably overlie the lower formations. Although the boundary between the Maekawa and the Haneo Formations is not exposed (Fig. 4), the Haneo Formation probably overlies the Maekawa Formation because of the nannofossil ages stated above. On the basis of benthic foraminiferal fossils and lithofacies, Yano [1986] estimated the paleodepth to be 200-600m (upper shelf slope) for the Maekawa and the Haneo Formations, 30-300m (sublittoral to lower shelf condition) for the Ninomiya Formation, and 30-100m (sublittoral to upper shelf condition) for the Numashiro Formation.

Many of the Pleistocene sediments in the Oiso Hills are tilted northward with dips of 5°-20° except in the Maekawa Formation, where dip directions of bedding planes diverge with a dip angle of at most 35° (Fig. 4).

The Oiso Hills are surrounded and cut along its margins by several active faults, such as the Kozu-Matsuda and the Shibusawa faults [Uesugi et al., 1982; Kano et al., 1984; Ito et al., 1987]. The Kozu-Matsuda fault along the southwestern margin of the Oiso Hills (Fig. 4) is known to have a large cumulative offset and a high rate of displacement. On the basis of the seismotectonics of the 1923 Kanto earthquake, the Kozu-Matsuda fault is estimated to be the upward extension of the inter-plate megathrust and to have a large right-lateral strike-slip component of displacement [Ando, 1971, 1974; Ishibashi, 1980]. Geologic evidence supports recent right-lateral offset across the Kozu-Matsuda fault [Uesugi et al., 1982]. The Kozu-Matsuda fault also has a reverse-slip component and the uplift of the Oiso Hills along the fault is estimated to have begun in the late Pleistocene at the latest [Yamazaki, 1984; Kano et al., 1984], or in the latest middle Pleistocene (0.3 Ma) [Ito et al., 1987]. The rate of the vertical displacement in the late Quaternary is estimated at 2-3.5m/1000 years [Matsushima, 1984; Yamazaki, 1984]. As for in the early middle Pleistocene and older ages, we have no evidence for the activity of the Kozu-Matsuda fault yet.

Samples and Measurement

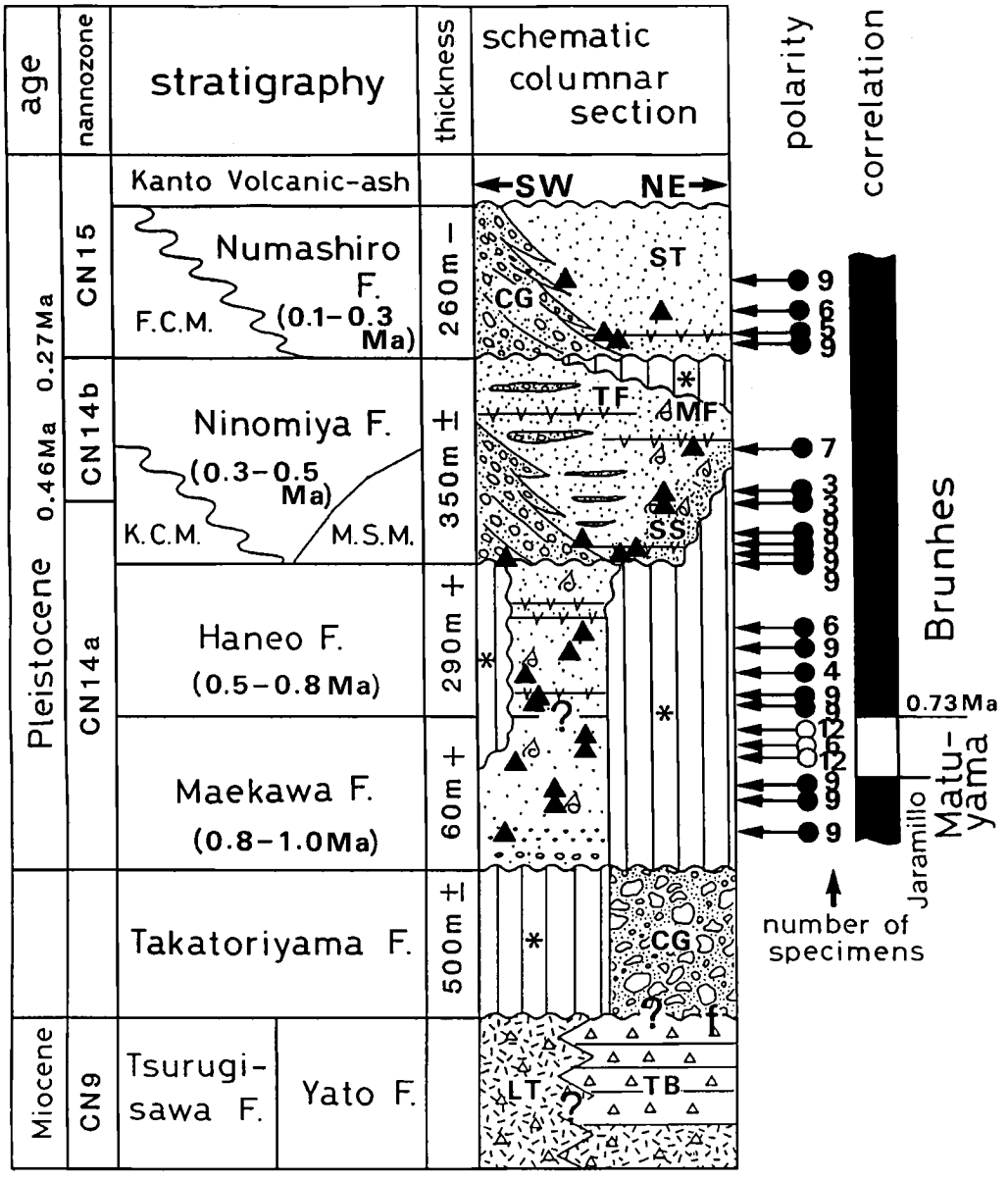
The paleomagnetic samples were collected from 52 horizons at 22 sites in the Maekawa, the Haneo, the Ninomiya, and the Numashiro Formations (Figs. 3 and 4, Table 1). One to three (mostly three) hand samples were collected from each sampling site. Each hand sample was collected from a different horizon. Each sampling horizon within the same site are 20cm to 4m apart vertically. All samples are fresh semi-consolidated bluish gray siltstones, sandy siltstones, or silty fine sandstones, which commonly have bioturbated features. The oriented block samples were drilled and cut into core specimens (24 mm in diameter and height) in the laboratory. Three to six specimens per horizon were measured for a total of 172 specimens.

All the paleomagnetic samples were measured and demagnetized at the Earthquake Research Institute, University of Tokyo. The remanent magnetizations of the samples were measured with a ring-core type spinner magnetometer [Koyama and Niitsuma, 1983]. All the specimens were subjected to alternating field (AF) demagnetizations in an apparatus with a three-axial set of demagnetizing coils in a magnetic shield [Koyama and Niitsuma, 1983]. At least one specimen from each horizon was subjected to stepwise AF demagnetizations up to 35-45 mT of peak field to test the stability of the magnetization. Stepwise thermal demagnetizations up to 560°-600°C were also carried out on selected pilot specimens. The thermal demagnetizer is composed of an electronic furnace in a magnetic shield and a temperature controller. The intensity of the magnetic field remained in the center of the furnace is around 20-30 nT.

Results

Alternating field (AF) demagnetization effectively removed secondary components of remanent

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F.C.M. : Fudosan Conglomerate Member
K.C.M. : Kozu Conglomerate Member
M.S.M. : Myoken Sandstone Member
F. : Formation

Fig. 3. Summary of stratigraphy, lithology, and magnetostratigraphy of the southern part of the Oiso Hills. Age ranges in parentheses in the column of stratigraphy are the nannofossil ages estimated by Okada [1987] (see text). In the column of schematic columnar section, solid triangles show the horizons of sampling sites. Rock patterns are CG, conglomerate; ST, siltstone; TF, key tuff bed; MF, molluscan fossils; SS, sandstone; LT, lapilli tuff; TB, tuff breccia. Pattern denoted by asterisk shows the lack of strata by unconformity. The magnetostratigraphic data were obtained in the present study. The other data are by Yano [1986].

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magnetization from all specimens. Most of the pilot specimens (from 47 horizons out of 52 in total) have strong stable components of remanent magnetization (Fig. 5). In these specimens, unstable components are completely removed by AF demagnetization of 25-30 mT or less. Thus, the remaining specimens from each horizon were demagnetized in 25-30 mT (Table 1). The other five pilot specimens (MK02A, MK02C, MK03C, MK07A, and NM02A in Table 1) have less stable magnetizations. In these specimens, regression line analysis was performed for the sublinear vector endpoints (at least three endpoints during 10-40 mT) on the Zijderveld diagrams, and the least-squares fit directions were calculated. All other specimens from these five horizons were also treated with stepwise AF demagnetizations and the same line-fitting procedures were performed.

Stepwise thermal demagnetizations were performed on four specimens of the Maekawa Formation and the results were compared with those of AF demagnetizations (Fig. 5). Stable endpoint magnetizations obtained by thermal treatment are nearly the same as those by AF demagnetization. This implies the absence of secondary magnetic minerals (e.g. goethite) having low blocking temperature and high coercive force, and suggests the validity of the AF demagnetizations to isolate stable directions of magnetization.

The magnetic properties of the present samples are summarized in Table 1. The intensity of natural remanent magnetization (NRM) of the specimens ranges from 1.0×10^{-5} to 1.6×10^{-4} kA/m (= emu/cc). The median demagnetizing field (MDF) of the specimens ranges from 6 to 40 mT (mostly 10 to 20 mT).

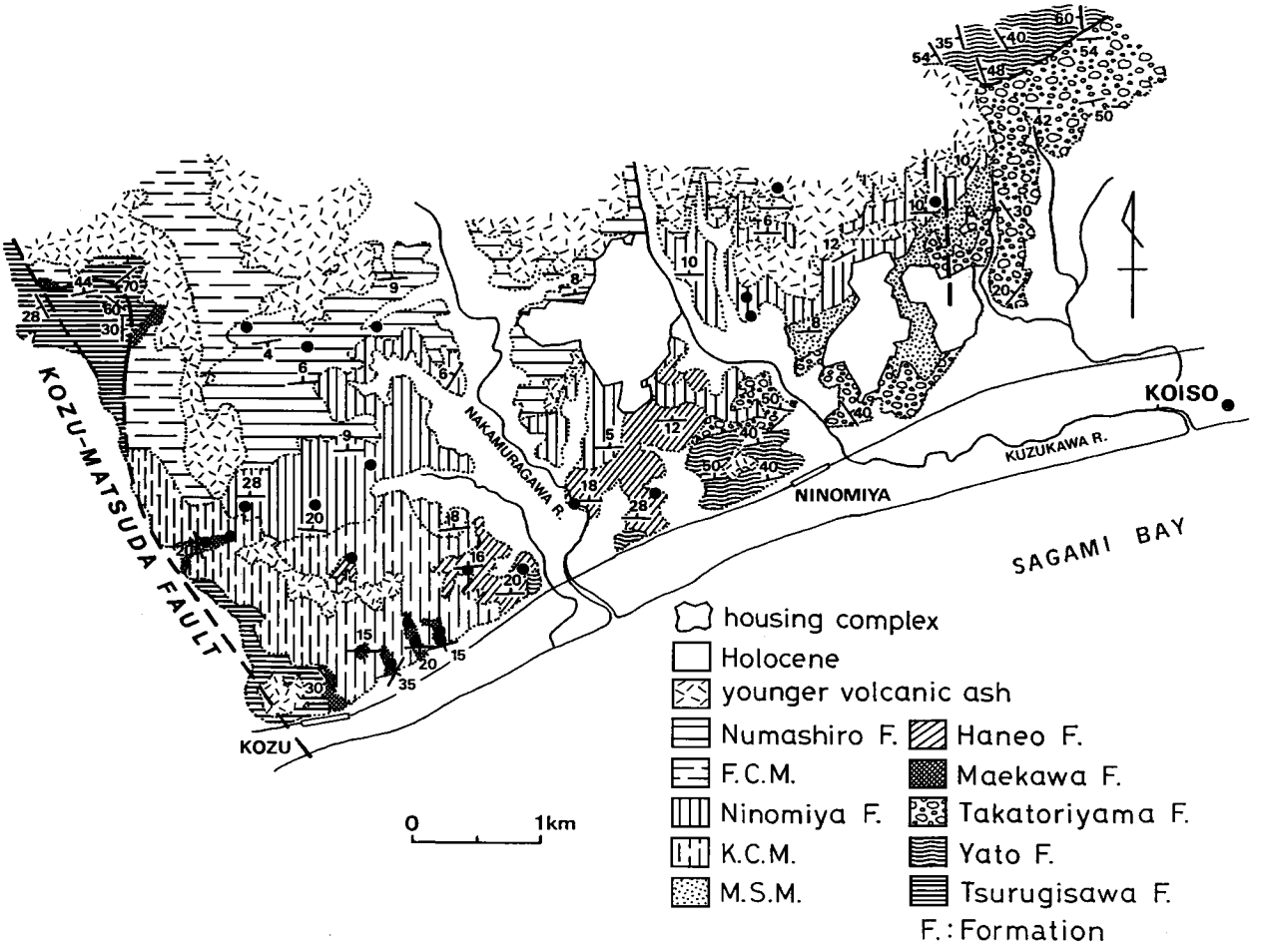


Fig. 4. Geologic map of the southern part of the Oiso Hills after Yano [1986]. F.C.M.: Fudosan Conglomerate Member. K.C.M.: Kozu Conglomerate Member. M.S.M.: Myoken Sandstone Member. Sampling sites for the present paleomagnetic study are shown with solid circles. KOISO: sampling site and results of the Nishikoiso Formation by Yoshida et al. [1984].

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TABLE 1. Paleomagnetic Results For Each Sampling Horizon In the Oiso Hills

horizon	N	MDF	AFD	Jn	D	I	Dc	Ic	k	α_{95}
Numashiro Formation										
NS01A	6	20	25	1.23×10-4	13.4	+61.3	13.4	+61.3	585	2.8
NS03A	5	20	25	9.83×10-5	14.9	+47.0	13.8	+43.2	889	2.6
NS04A	3	17	25	9.25×10-5	346.3	+40.4	347.4	+34.6	1123	3.7
NS04B	3	20	25	1.33×10-4	351.1	+32.6	351.7	+26.7	4559	1.8
NS04C	3	17	25	8.42×10-5	1.1	+45.1	1.1	+39.1	1668	3.0
NS05A	3	14	25	5.17×10-5	355.5	+34.1	1.6	+40.0	997	3.9
NS05B	3	10	25	4.05×10-5	4.6	+46.1	14.8	+50.2	79.2	13.9
NS05C	3	15	25	6.34×10-5	357.4	+38.6	4.8	+44.0	475	5.7
Ninomiya Formation										
NM01A	3	15	25	9.72×10-5	15.0	+65.9	3.7	+60.3	1993	2.8
NM01B	3	17	25	8.36×10-5	3.4	+55.9	357.2	+49.4	786	4.4
NM01C	3	18	25	8.95×10-5	6.1	+58.2	358.9	+51.9	1111	3.7
NM02A	3	17	15-40	1.01×10-4	358.8	+63.0	4.0	+55.6	179	9.3
NM02B	3	14	25	7.27×10-5	5.7	+58.7	8.9	+51.1	1413	3.3
NM03B	3	16	25	6.45×10-5	17.9	+59.2	14.3	+52.6	310	7.0
NM03C	4	20	25	5.75×10-5	21.3	+54.8	17.8	+48.4	577	3.8
KZ01A	3	20	25	1.43×10-4	6.1	+72.1	20.0	+40.1	2456	2.5
KZ01B	3	18	25	1.63×10-4	1.2	+71.2	17.7	+39.7	1576	3.1
KZ01C	3	16	25	1.07×10-4	349.7	+66.4	10.5	+36.8	583	5.1
NM04A	3	18	25	9.27×10-5	338.3	+80.0	351.2	+50.3	244	7.9
NM04B	3	18	25	9.70×10-5	302.7	+76.9	339.2	+50.9	434	5.9
NM04C	3	14	25	7.22×10-5	2.7	+71.3	358.9	+41.4	701	4.7
NM05A	3	20	25	9.54×10-5	350.8	+59.3	356.4	+42.9	1113	3.7
NM05B	3	13	25	7.88×10-5	336.4	+63.4	347.8	+48.1	2689	2.4
NM05C	3	13	25	9.22×10-5	357.8	+64.2	1.8	+47.4	2100	2.7
Haneo Formation										
HN01A	3	9	25	1.59×10-5	356.9	+75.3	353.1	+55.4	773	4.4
HN01B	3	10	25	1.95×10-5	353.1	+67.4	351.8	+47.4	161	9.7
HN01C	3	9	25	1.48×10-5	9.6	+62.7	2.2	+43.4	141	10.4
HN02A	3	11	25	4.07×10-5	21.6	+67.9	19.4	+52.0	427	6.0
HN02B	3	6	25	3.24×10-5	357.2	+72.1	5.6	+56.6	259	7.7
HN02C	3	11	25	3.19×10-5	15.7	+64.9	15.8	+48.9	229	8.2
HN03A	6	12	25	1.49×10-5	21.4	+48.3	25.6	+39.3	65.9	8.3
HN05A	4	13	25	1.12×10-5	0.9	+52.1	8.2	+44.4	220	6.2
HN06A	3	16	30	7.97×10-5	25.9	+72.4	17.4	+61.0	433	5.9
HN06B	3	22	30	1.34×10-4	6.1	+48.6	5.8	+36.6	1285	3.4
HN06C	3	19	30	1.19×10-4	22.9	+70.7	16.0	+59.1	569	5.2
Maekawa Formation										
MK01A	3	18	25	1.16×10-4	55.5	+40.0	73.5	+45.1	659	4.8
MK01B	3	20	25	1.13×10-4	53.9	+22.6	63.2	+29.7	166	9.6
MK01C	3	18	25	1.26×10-4	43.5	+30.7	56.2	+40.6	1256	3.5
MK02A	3	11	10-35	1.54×10-5	26.9	+50.7	26.9	+50.7	771	4.4
MK02B	3	12	25	1.85×10-5	53.0	+56.3	53.0	+56.3	77.2	14.1
MK02C	3	8	10-30	1.01×10-5	6.4	+57.2	6.4	+57.2	138	10.5
MK03A	4	40	30	5.19×10-5	182.7	-33.2	209.8	-41.4	259	5.7
MK03B	4	40	30	4.64×10-5	174.6	-37.4	207.2	-49.0	880	3.1
MK03C	4	22	20-35	1.36×10-4	193.6	-45.2	229.4	-43.5	132	8.0
MK04A	4	35	30	4.01×10-5	234.5	-45.8	250.6	-48.8	507	4.1
MK04B	4	25	30	3.79×10-5	247.8	-22.1	254.0	-22.9	192	6.7
MK04C	4	32	30	2.86×10-5	229.4	-34.8	240.3	-39.7	229	6.1
MK05A	3	13	25	6.83×10-5	63.5	+55.6	70.5	+44.0	216	8.4
MK05B	3	12	25	3.80×10-5	65.7	+63.1	74.0	+51.1	350	6.6
MK05C	3	12	25	7.13×10-5	48.2	+63.3	61.6	+53.1	286	7.3
MK07A	3	10	20-40	5.07×10-5	202.8	-48.7	202.8	-48.7	108	11.9
MK07B	3	31	30	5.99×10-5	218.0	-40.9	218.0	-40.9	109	11.8

N: number of specimens. MDF: median demagnetizing field in mT. AFD: adopted intensity of alternating field demagnetization in mT (see text). Jn: intensity of natural remanent magnetization in kA/m (= emu/cc). D and I: declination and inclination of remanent magnetization before tilting correction. Dc and Ic: declination and inclination of remanent magnetization after tilting correction. k: precision parameter. α_{95}: 95% confidence limit.

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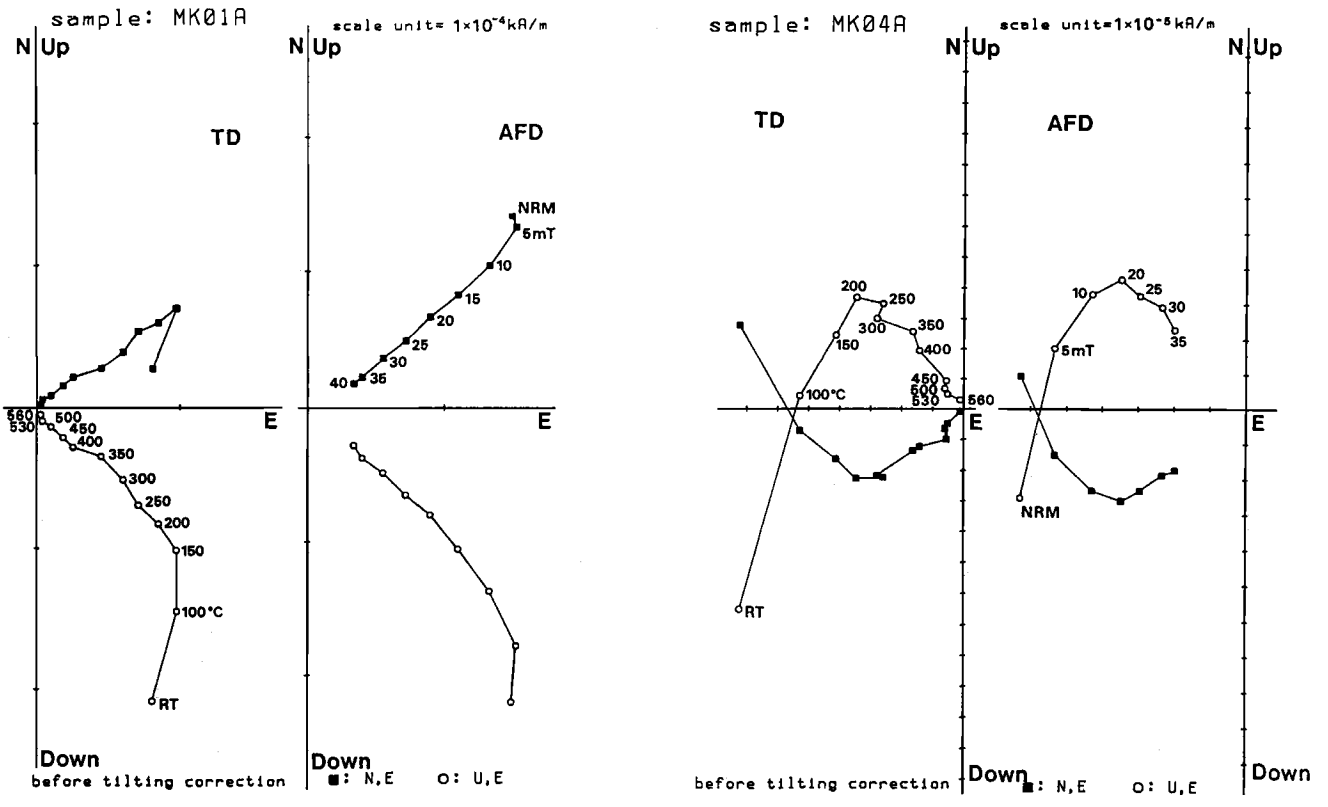


Fig. 5. Comparison between the Zijderveld diagrams from stepwise alternating field demagnetizations and those from stepwise thermal demagnetizations. Each pair of specimens is collected from the same horizon. TD: results by thermal demagnetizations. RT: room temperature. AFD: results by alternating field demagnetizations. NRM: natural remanent magnetization.

Although the fold test is not applicable because of the low dip angle of strata at the sampling sites (Fig. 4), the remanence directions from the Maekawa Formation have both normal and reversed polarities and show a nearly antipodal distribution (Fig. 6, Table 2). Moreover, the present samples are fresh and semi-consolidated, without any weathering, alteration, deformation or heavy diagenesis. These facts suggest that any unremoved secondary components in these samples are negligibly small, and that the stable directions after AF demagnetization (Fig. 6) are probably the primary records of paleo-geomagnetic field.

The magnetic polarities of all the sampling sites and the correlation to the standard magnetic polarity time scale [Mankinen and Dalrymple, 1979] are shown in Fig. 3. Out of the 22 sampling sites, the 16 sampling sites in the Numashiro, the Ninomiya, and the Haneo Formations show normal polarities, and the six sampling sites in the Maekawa Formation show both normal and reversed polarities (Figs. 3 and 6).

Since the age of the Maekawa Formation is estimated to be 0.8-1.0 Ma from calcareous nannofossil biochronology (Section 2), we correlate the normal and the reversed polarities of the Maekawa Formation to the Jaramillo normal polarity subchron and the Matuyama reversed polarity chron, respectively (Fig. 3). The normal polarity interval over the Haneo, the Ninomiya, and the Numashiro Formations is correlated to the Brunhes normal polarity chron (Fig. 3) also on the basis of the nannofossil ages.

Tectonic Movement

Tectonic Significance of Paleomagnetic Direction

The mean field direction and its confidence limit in each sampling horizon are shown in Fig. 6 and Table 1, and those in each formation are summarized in Table 2. The mean direction of remanence declination in each site is shown in Fig. 7.

The mean directions from the Numashiro, the Ninomiya, and the Haneo Formations are nearly aligned with the present axial geocentric dipole field (D= 0°, I= +54.3°) (Fig. 6, Table 2). On the

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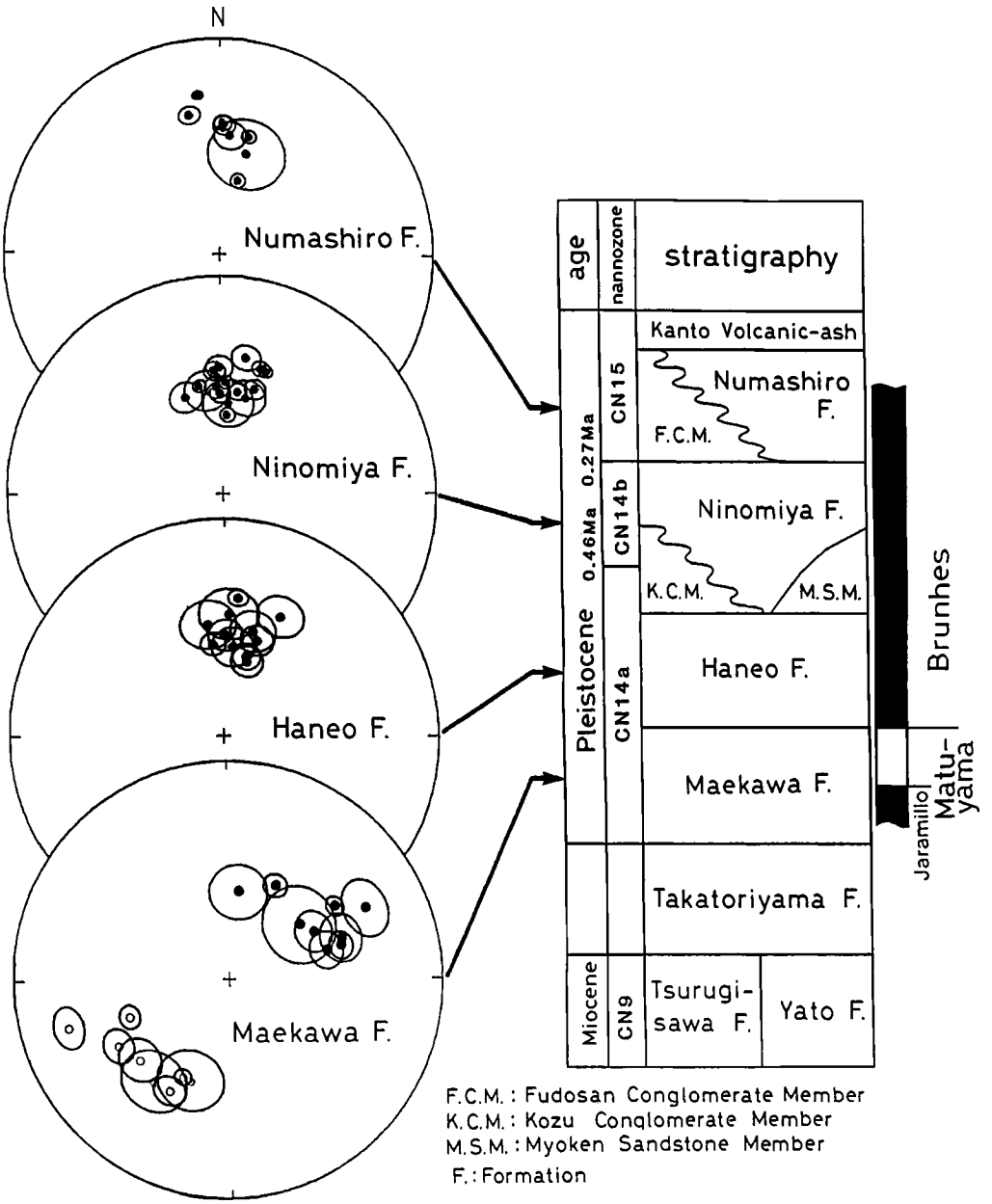


Fig. 6. Horizon-mean directions of remanent magnetization after demagnetization and tilting correction. All directions are plotted on stereonets of equal area projection. Solid circle: direction on lower hemisphere. Open circle: direction on upper hemisphere. Circle around each direction shows the 95% confidence limit.

basis of the magnetostratigraphy (Section 4) and the calcareous nannnofossil ages (Section 2), the sampling horizons from these three formations range in age from 0.2 to 0.7 Ma. The time interval sampled is sufficiently long to ensure that the geomagnetic secular variation averages to a geocentric axial dipole field. Thus, it is concluded that the Numashiro, the Ninomiya, and the Haneo Formations have undergone no tectonic movements detectable by paleomagnetism.

In contrast, the mean directions from the Maekawa Formation show significant clockwise

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TABLE 2. Mean Remanence Direction For Each Stratigraphic Unit After Tilting Correction

Formation	N	n	Dc	Ic	k	α_{95}	PHI	LMD	A_{95}
Numashiro	8	29	2.4	+42.8	40.3	8.8	80.0	-57.1	8.7
Ninomiya	16	49	3.4	+48.5	66.0	4.6	84.0	-68.1	5.3
Haneo	11	37	9.1	+49.9	57.5	6.1	81.5	-106.4	6.7
Maekawa(normal)	9	27	56.1	+49.4	23.8	10.8			
Maekawa(reversed)	8	30	227.6	-43.5	22.6	11.9			
Maekawa(total)	17	57	51.8	+46.7	23.2	7.6	46.5	-136.9	8.9

N: number of sampling horizons. n: number of specimens. Dc and Ic: declination and inclination of mean remanence direction after tilting correction. k: precision parameter. α_{95}: 95% confidence limit. PHI, LMD, A_{95}: north latitude, east longitude, and 95% confidence limit of virtual geomagnetic pole (north-seeking pole).

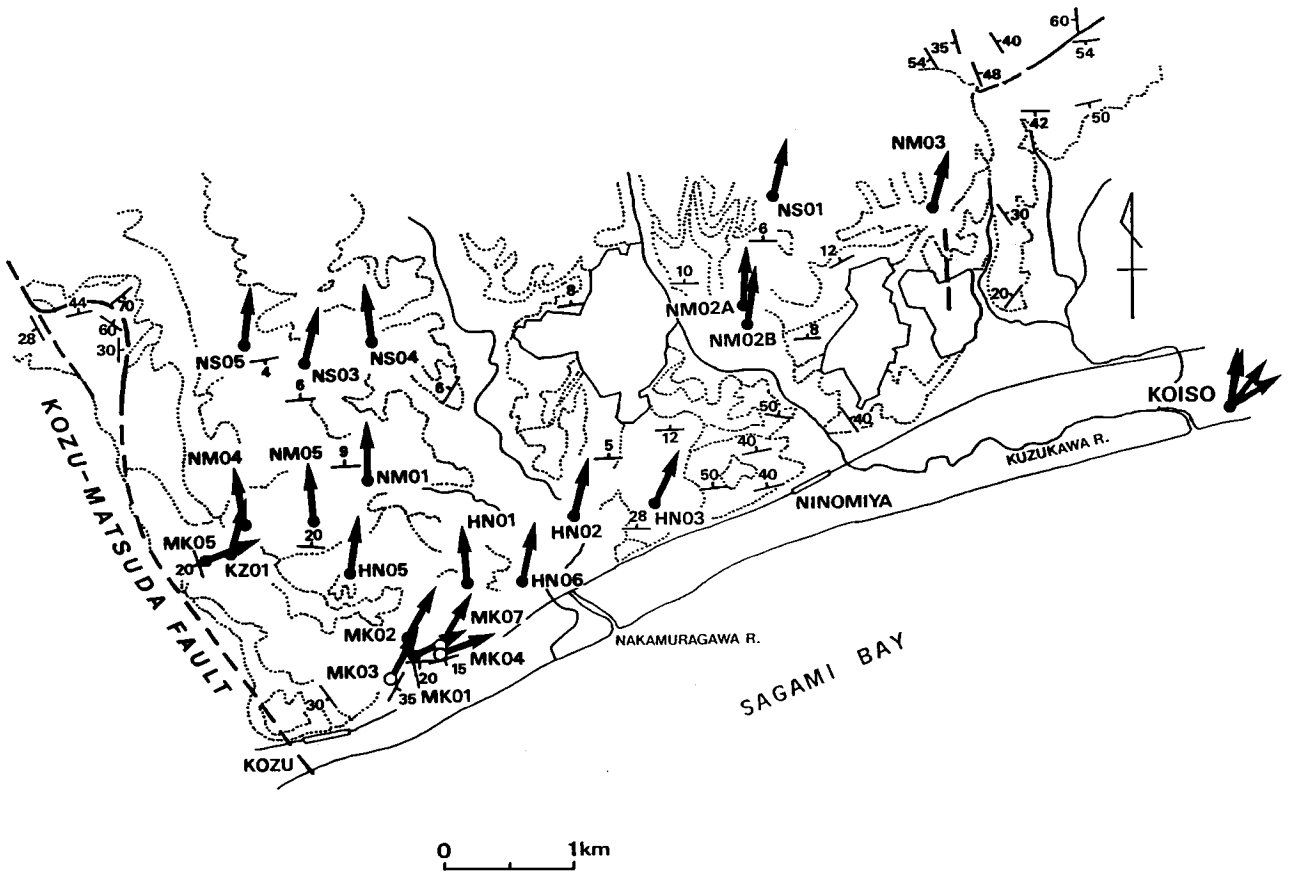


Fig. 7. Map showing the mean direction of declination of remanent magnetization in each site. All the declinations were converted to normal polarity for the sake of visual simplicity. Solid circle: site of normal polarity. Open circle: site of reversed polarity. KOISO: sampling site and results of the Nishikoiso Formation by Yoshida et al. [1984].

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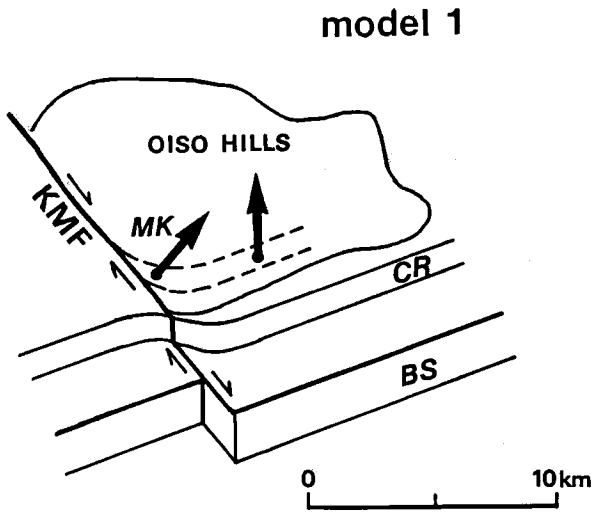


Fig. 8. Schematic diagram showing the clockwise rotation of paleomagnetic vectors caused by local right-lateral strike-slip drag along the Kozu-Matsuda fault (model 1, see text). KMF: Kozu-Matsuda fault. MK: direction of paleomagnetic declination of the Maekawa Formation. CR: post-Miocene strata. BS: basement of older rocks. Broken lines show the trace of strike of the

deflections of declination (52° ±11° on average) (Fig. 6, Table 2). The antipodal distribution of the directions of both normal and reversed polarities suggests that the clockwise deflection is not a reflection of short term geomagnetic secular variation or a polarity transition. Miscorrection for polyphase tilting also cannot explain such large deflections of declination, because the dip angles of strata at the sampling sites of the Maekawa Formation are at most 35° (Section 2) [MacDonald, 1980].

Therefore, we conclude that the declination deflections of the Maekawa Formation was caused by the clockwise tectonic rotation of the sampling sites. The paleomagnetic results (Fig. 6) show that the tectonic rotation occurred between the latest stage of the deposition of the Maekawa Formation and the earliest stage of that of the Haneo Formation, that is between 0.6 and 0.9 Ma.

Model of Tectonic Rotation

What are the mechanism and the cause of the tectonic rotation recorded in the Maekawa Formation? We propose the following models 1 to 3, which attribute the tectonic rotation to (1) local drag on a major strike-slip fault, to (2) regional dextral shear along a mechanical boundary zone between plates, and to (3) oroclinal bending of the study region, respectively.

Model 1 attributes the tectonic rotation of the Maekawa Formation to a local deformation by the right-lateral drag along the Kozu-Matsuda fault (Fig. 8), and does not regard the rotation as a part of regional deformation caused by a common mechanism. Drag rotations of paleomagnetic directions are expected to exist along a fault with a large strike-slip component [MacDonald, 1980]. In this case, clockwise drag rotaion is accompanied with a right-lateral strike-slip displacement, while anticlockwise one is accompanied with a left-lateral one. Since the Kozu-Matsuda fault is thought to have a large right-lateral strike-slip component (Section 2) and the Maekawa Formation is distributed near the Kozu-Matsuda fault (Figs. 4 and 7), this model may explains the tectonic rotation of the Maekawa Formation.

Model 2 regards the tectonic rotation of the Maekawa Formation as a part of the regional dextral shear deformation along the mechanical boundary zone between the Philippine Sea (PHS) plate and northeast Japan (Fig. 9). Many examples of tectonic rotation in a highly faulted areas are revealed by paleomagnetism [e.g. Luyendyk et al., 1985; Wells and Coe, 1985; Koyama, 1983] or by seismology [Nicholson et al., 1986a, 1986b]. A right-lateral strike-slip component of displacement is expected to be dominant in the mechanical boundary zone because of the plate geometry (Fig. 2). Thus, in the Oiso Hills, clockwise block rotations may commonly occur along the mechanical boundary zone in response to regional dextral shear.

Model 3 regards the tectonic rotation in the Maekawa Formation as a part of the northward oroclinal bending of the south Kanto region (Fig. 10). Yoshida et al. [1984] proposed the clockwise rotation of the Miura Peninsula of approximately 30° since 4 Ma on the basis of paleomagnetism (Fig. 10). Ota et al. [1986] documented by paleomagnetism the clockwise rotation (40° on average) of the northeastern Tanzawa Mountains since the late Miocene (Fig. 10). Koyama and Kotake [1987] carried out the paleomagnetic study of the southern Boso Peninsula and estimated the clockwise tectonic rotation (12° on average) since 1 Ma (Fig. 10). All these data suggest that the clockwise tectonic rotations occurred not only in the Oiso Hills but also in the wider area of the south Kanto region.

The reality of model 1 depends on whether the Kozu-Matsuda fault existed or not during the time of the tectonic rotation of the Maekawa Formation, and on whether drag rotation has actually occurred or not during the time of the latest activity of the fault since 0.3 Ma (Section 2). If model 1 is the case, the drag deformation along the Kozu-Matsuda fault should have accumulated for the last 0.3 million years. We have no evidence for the activity of the fault during the time of the tectonic rotation (0.6-0.9 Ma) (Section 2). Moreover, any site of the Haneo, the Ninomiya, and the Numashiro Formations shows no significant clock-

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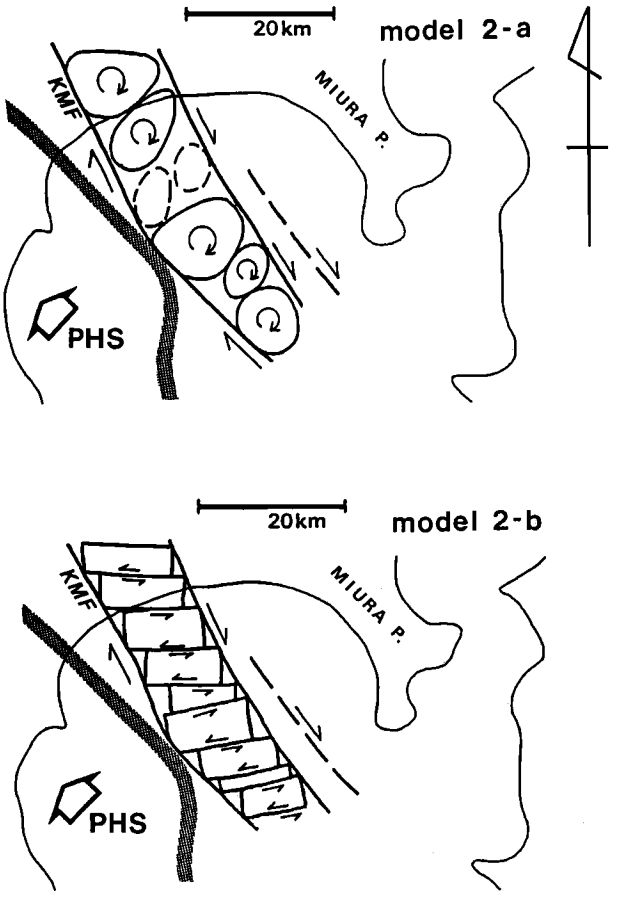


Fig. 9. Schematic diagrams showing possible regional dextral shear in the early Quaternary along the mechanical boundary zone between the Philippine Sea (PHS) plate and northeast Japan (model 2, see text). Shaded belt: material boundary between the plates. Large arrow: estimated convergence direction between the PHS plate and northeast Japan in the early Quaternary before 0.5 Ma [Seno, 1985]. Thick lines show major faults or margins of faulted blocks comprising the mechanical boundary zone between the plates. These are estimated from the geomorphology and the fault distribution in Sagami Bay (Fig. 13). Small arrow: sense of fault displacement. Thin line: coastline at the present. KMF: Kozu-Matsuda fault. MIURA P.: Miura Peninsula. In this model, alternative two mechanisms can explain the tectonic rotation. Model 2-a (upper figure) is based on the "ball bearing" model by Beck [1976], where the crustal blocks located between a set of right-lateral strike-slip faults are clockwise rotated by drag along the faults. Model 2-b (lower figure) is based on the "Riedel shear rotation" model by Freund [1974], where elongate crustal slivers and the left-lateral local faults bordering the slivers are clockwise rotated together.

wise deflections near the Kozu-Matsuda fault (e.g. KZ01, NM04, NM05, HN05, or NS05 in Fig. 7). This suggests that no large drag rotation along the Kozu-Matsuda fault has occurred since the time of deposition of the Haneo Formation.

The well-constrained age range (0.6-0.9 Ma) of the tectonic rotation of the Maekawa Formation means that the rotation started and ended within a rather short interval. We think that some regional tectonic event is required to explain such a rapid and large deformation. As stated below, at least two possible tectonic events are known to exist in the early Quaternary and they can explain the tectonic rotation of the Maekawa Formation. Thus, model 2 or 3, which assumes a regional deformation, is more likely to explain the tectonic rotation than model 1.

The paleomagnetic data from the Miocene basement in the Oiso Hills show clockwise deflections of declination of 8°-59° [Yoshida et al., 1984, Fig. 11]. These samples were collected from the four horizons in the late Miocene Nishikoiso Formation on the east side of the study area (the site KOISO in Figs. 4 and 7). Although this result is based on the data from only four sampling horizons, the clockwise deflections of declination are similar to the present result from the Maekawa Formation (Figs. 7 and 11). Hence, it is possible that the deflections of declination both in the Nishikoiso and in the Maekawa Formations, located 6 km apart, were caused by the common mechanism of tectonic rotation affecting the whole Oiso Hills, suggesting the validity of model 2 or 3.

The rotation age of the Nishikoiso Formation is, however, not well constrained; this age is estimated at sometime between the late Miocene and the present on the basis of the age of foraminiferal fossils [Ibaraki, 1978]. If the rotation of the Nishikoiso Formation occurred prior to that of the Maekawa Formation, a different mechanism of tectonic rotation is possible for the Nishikoiso Formation.

Similarly, the reality of model 3 depends on the ages of the tectonic rotations in other areas in the south Kanto region. As stated above, these ages have not been so well constrained as for the Maekawa Formation. If the ages of these rotations are coeval with the rotation of the Maekawa Formation (0.6-0.9 Ma), all these clockwise tectonic rotations are likely to have been caused by the same mechanism, that is of model 3. If not so, a different mechanism of tectonic rotation is possible for each area.

What is the cause of the tectonic rotation in models 2 and 3? We point out and discuss the following two possible tectonic events, (1) collision and indentation of buoyant terranes or (2) change of relative motion between plates.

[scenario 1]: collision and indentation of buoyant terranes

The Izu block is thought to have collided with Honshu in the early Quaternary (Section 1). The

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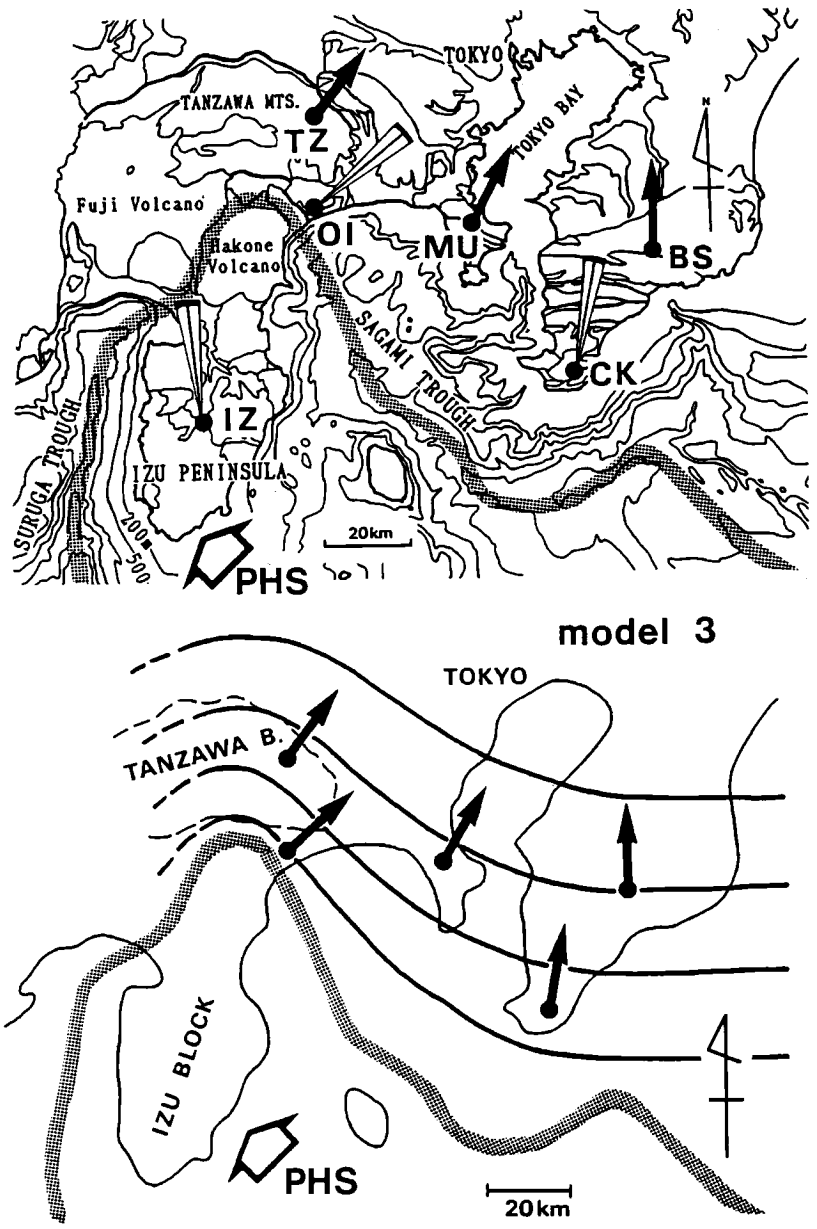


Fig. 10. Upper figure is a map showing the directions of paleomagnetic declination obtained in the Oiso Hills and adjacent areas in the south Kanto region. Shaded belt: material boundary between plates [Nakamura et al., 1984]. Large arrow: estimated convergence direction between the Philippine Sea plate and northeast Japan in the early Quaternary before 0.5 Ma [Seno, 1985]. OI: Maekawa Formation in the Oiso Hills (this study). MU: Miura Peninsula [Yoshida et al., 1984]. TZ: northeastern Tanzawa Mountains [Ota et al., 1986]. CK: Chikura Group [Koyama and Kotake, 1987]. IZ: upper part of the Shirahama Group in the Izu Peninsula [Koyama, 1986b]. BS: Boso Peninsula [Niitsuma, 1976]. The ages of the samples are late Pliocene to early Pleistocene (OI, CK, IZ, and BS), and middle Miocene to early Pliocene (TZ and MU). For further details of each result, the reader may refer to text and Koyama [1988]. Lower figure is a schematic diagram showing possible northward oroclinal bending of the south Kanto region caused by the collision and indentation of buoyant terranes (Izu block and/or Tanzawa block) against Honshu (model 3, see text). The mode of the bending is schematically shown by thick lines and arrows. These arrows are the observed directions of paleomagnetic declination shown in the upper figure. Thin solid line: coastline at the present. Thin broken line: location of the Tanzawa block. Other symbols are the same as in the upper figure.

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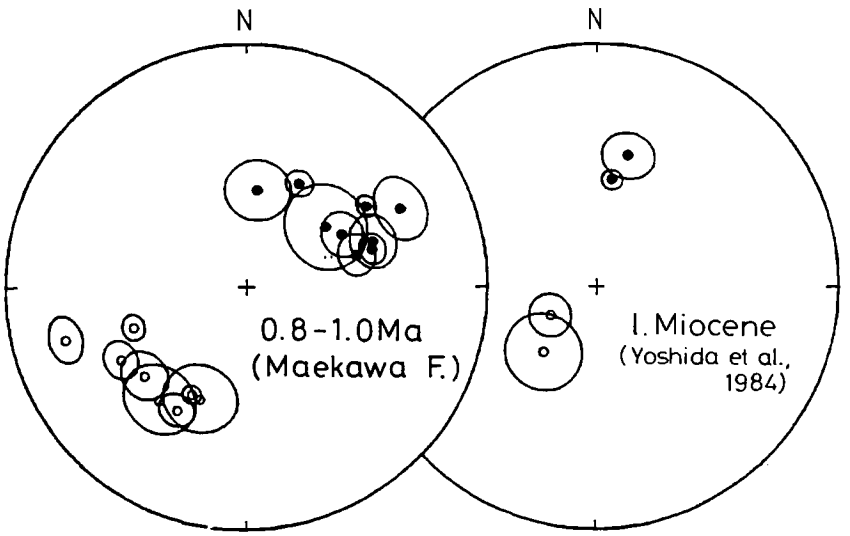
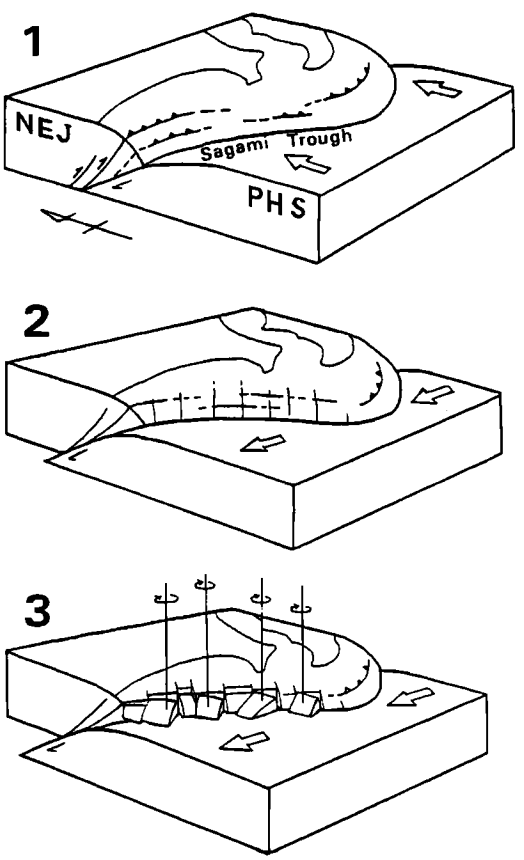


Fig. 11. Comparison between the directions of remanent magnetization from the Maekawa Formation (left circle) and those from the Nishikoiso Formation by Yoshida et al. [1984] (right circle). All directions are plotted on stereonets of equal area projection. Solid circle: direction on a lower hemisphere. Open circle: direction on a upper hemisphere. Circle around each direction shows the 95% confidence limit.



age of the collision is estimated to be 0.7 to 1.0 Ma on the basis of geology and benthic foraminiferal paleobathymetry in the Ashigara and other areas around the Izu block [Huchon and Kitazato, 1984; Kitazato, 1986]. The Izu block is regarded as a buoyant terrane which have indented the Honshu arc [Matsuda, 1978; Ishibashi, 1986]. Thus, the areas on the northeast of the Izu block were possibly pushed aside and rotated clockwise (model 3, Fig. 10) as proposed by Matsuda [1984], Yoshida et al. [1984], and Ishibashi [1986], or possibly underwent regional dextral shear (model 2, Fig. 9) by the drag along the margin of the buoyant Izu block.

The Ashigara Group, which is distributed in the Ashigara area on the northwest of the Oiso Hills (Fig. 2), was strongly deformed during the interval from 0.5 to 0.8 Ma, possibly the result of indentation by the colliding Izu block [Koyama, 1986a]. Since this deformation age agrees well

Fig. 12. Schematic diagram showing the possible tectonic rotations caused by the eduction of the Philippine Sea plate (see text). NEJ: northeast Japan. PHS: Philippine Sea plate. The convergence direction of the Philippine Sea plate (large arrow) changed from northward (upper figure) to northwestward (middle figure), and disintegration of the continental slope was started by the drag along the eduction boundary (middle figure). As a result, rearrangement and rotation of blocks occurred (lower figure).

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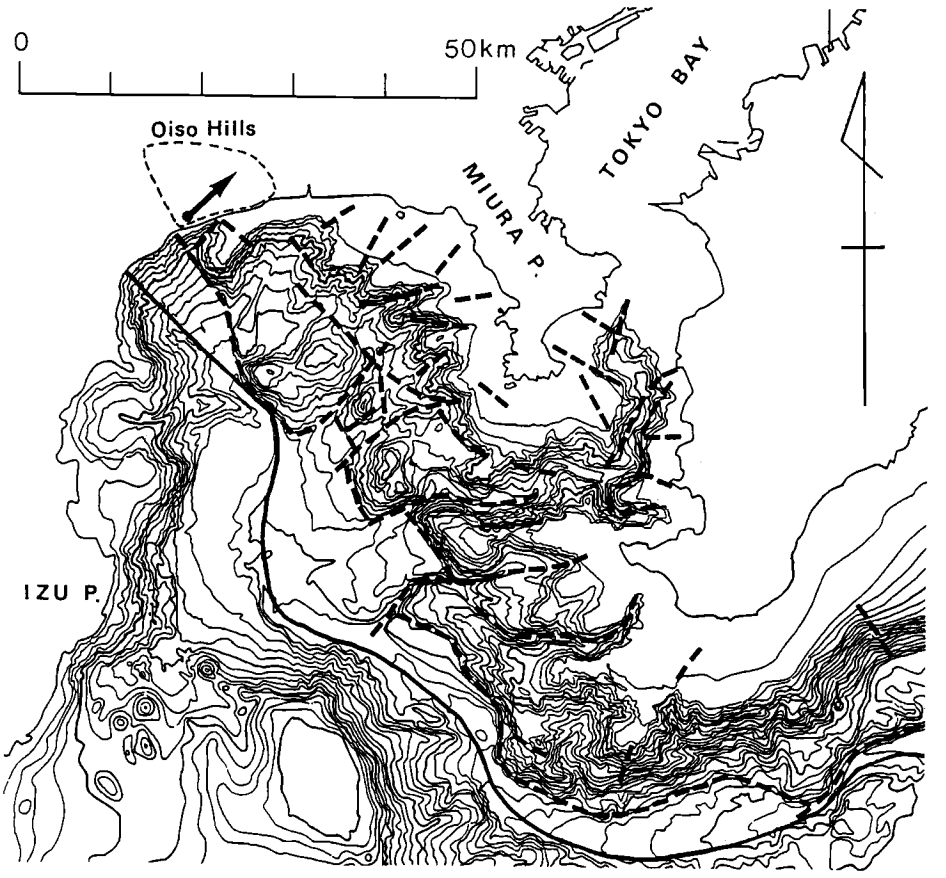


Fig. 13. Bathymetric chart of Sagami Bay [Hydrographic Department, M.S.A., 1985]. Depth contour interval is 100m. Thick broken lines show the distribution of major submarine faults [Kimura, 1976]. Thick solid line: material boundary between plates by Nakamura et al. [1984]. The Oiso Hills and the direction of the mean declination of the Maekawa Formation obtained in the present study are also shown. IZU P. and MIURA P.: Izu and Miura Peninsulas.

with that of the tectonic rotation of the Maekawa Formation (0.6-0.9 Ma), the rotation of the Maekawa Formation can be explained by the collision of the Izu block with Honshu.

The Tanzawa Mountains, which are located to the north of the Izu block (Fig. 2), are regarded as an allochthonous terrane that accreted at about 5 Ma prior to the collision of the Izu block with Honshu [Niitsuma and Matsuda, 1985]. Thus, as in the case of the Izu block, the collision and indentation of the Tanzawa block possibly caused tectonic rotations in the south Kanto region. If the age of the tectonic rotation of the Nishikoiso Formation or of the Miura Peninsula stated above is significantly older than that of the Maekawa Formation, they may be related to the collision and indentation of the Tanzawa block.

[scenario 2]: change of relative motion between plates

On the basis of plate geometry, geology, and geomorphology in and around the study region, Nakamura et al. [1984] and Kaizuka [1984] proposed that the relative convergence orientation between the Philippine Sea (PHS) and the Eurasian plates at about 1 Ma was closer to a northward azimuth than the present northwestward direction. Nakamura et al. [1984] thought that this change transformed a subduction boundary in the Sagami Bay region into an eduction boundary, where the once subducted PHS plate emerging (educting) from beneath the northeast Japan arc (Fig. 12). In Sagami Bay, there are a row of submarine banks along the northeastern slope of the Sagami trough (Fig. 13). The Oiso Hills are thought to be the northernmost subaerial bank [Kaneko, 1971]. Nakamura et al. [1984] thought that these banks and the curved canyons associated with them are the fault-related topography reflecting the eduction of the PHS plate, and that they were produced by the disintegration and rearrangement of the blocks which once composed a smooth continental slope (Fig. 12). Such disintegration and rearrangement may

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have accompanied large-scale rotations of the blocks, and the Oiso Hills, one of these blocks, may have been rotated by this mechanism (Fig. 12).

In this case, large deformations are expected only along the mechanical boundary zone between the plates, and thus model 2 is valid to explain the tectonic rotation of the Maekawa Formation.

On the basis of geomorphology around the triple junction among the plates to the southeast of the study region (Fig. 1), Seno [1987] inferred that the direction of the relative motion between the PHS plate and northeast Japan changed at about 1.5 Ma, and the eduction in the Sagami Bay region had begun and continued up to 0.5 Ma. If this is the case, model 2 seems to be improbable, because the age range of the eduction (0.5-1.5 Ma) is much longer than that of the rotation of the Maekawa Formation (0.6-0.9 Ma).

In contrast with Seno [1987], Ishibashi [1986] doubted the change of the relative motion between the PHS plate and northeast Japan in the early Quaternary. He thought that the geomorphology of the continental slope in Sagami Bay arose from the right-lateral strike-slip drags caused by the indentation of the buoyant Izu block in the early Quaternary.

The abrupt initiation and the high deformation rate of the tectonic rotation of the Maekawa Formation can be explained by model 2 or 3. However, we think that the reason for the abrupt cessation of the rotation is unclear yet. Such abrupt cessation of tectonic rotation may occur when the rotation angle attains to some mechanical limit.

Conclusion

A part of the Oiso Hills (the Maekawa Formation) underwent clockwise tectonic rotation of about 50° during the interval from 0.6 to 0.9 Ma. This rapid large rotation was probably caused by regional dextral shear in the mechanical boundary zone between the Philippine Sea plate and northeast Japan, or by northward oroclinal bending of the whole south Kanto region associated with the collision and indentation of the Izu block against Honshu. The regional dextral shear is probably associated also with the collision and indentation of the Izu block, or with the change of relative convergence direction between the plates. It is difficult at present, however, to choose among the models of rotation because we have an insufficient number of observations. We need to collect more geologic and paleomagnetic data from the Oiso Hills and adjacent areas including the Sagami Bay region to define better the rotated domains.

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